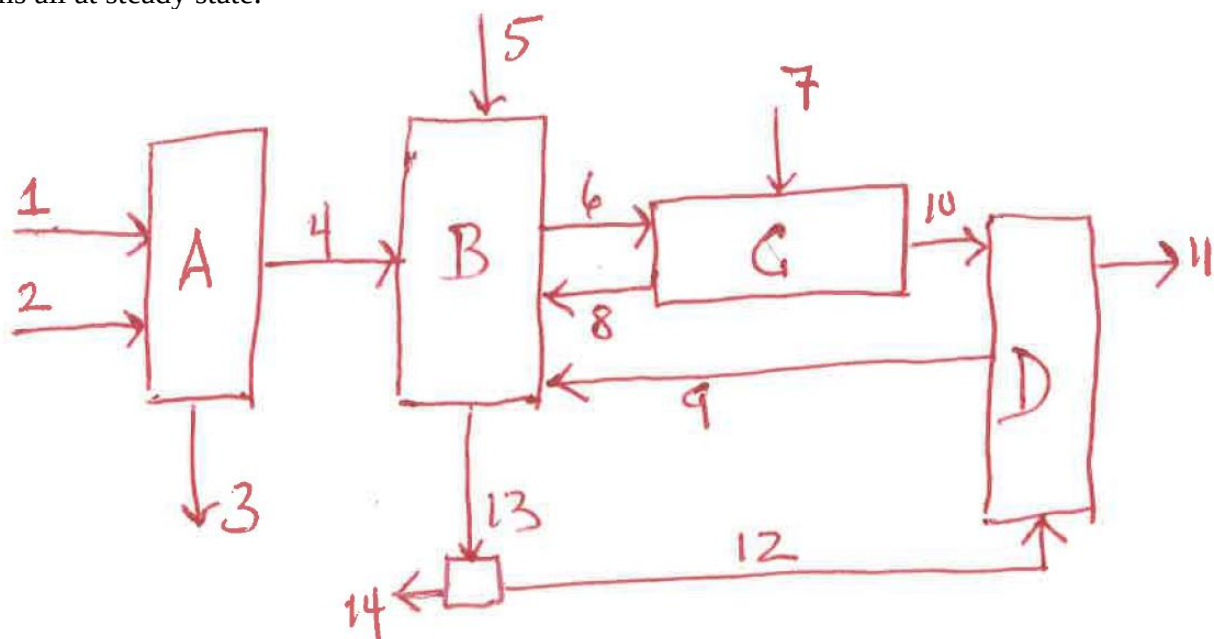


Short Answers (35 points total)

1. (5 pts.) List the four types of process units found in block flow diagrams.
2. (10 pts.) Four separators, labeled A, B, C, and D and one splitter (shown below) are connected by fourteen streams all at steady state.



Complete the following table of total mass flow rates for the fourteen streams by filling the five missing entries:

Stream No.	1	2	3	4	5	6	7
Flow Rate (kg/h)	45		32	238		244	26

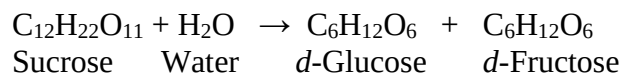
Stream No.	8	9	10	11	12	13	14
Flow Rate (kg/h)	80			187	252		120

3. (20 pts.) Fuel oil is burned with air. The flue gas exiting the burner contains 10.2 mol% CO₂, 8.3 mol% O₂, and the remainder N₂ **on a dry basis**. The fuel oil is made up of molecules containing only carbon and hydrogen atoms.
- (a) What is the molar ratio of carbon to hydrogen in the fuel oil?
 - (b) What is the percentage excess air fed to the burner?

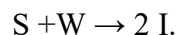
Open Book and Closed Notes

SPECIAL NOTE: Do all calculations in the exam booklet provided. To receive full credit, write your final answer in the space provide on this sheet. You must turn this sheet with your exam booklet.

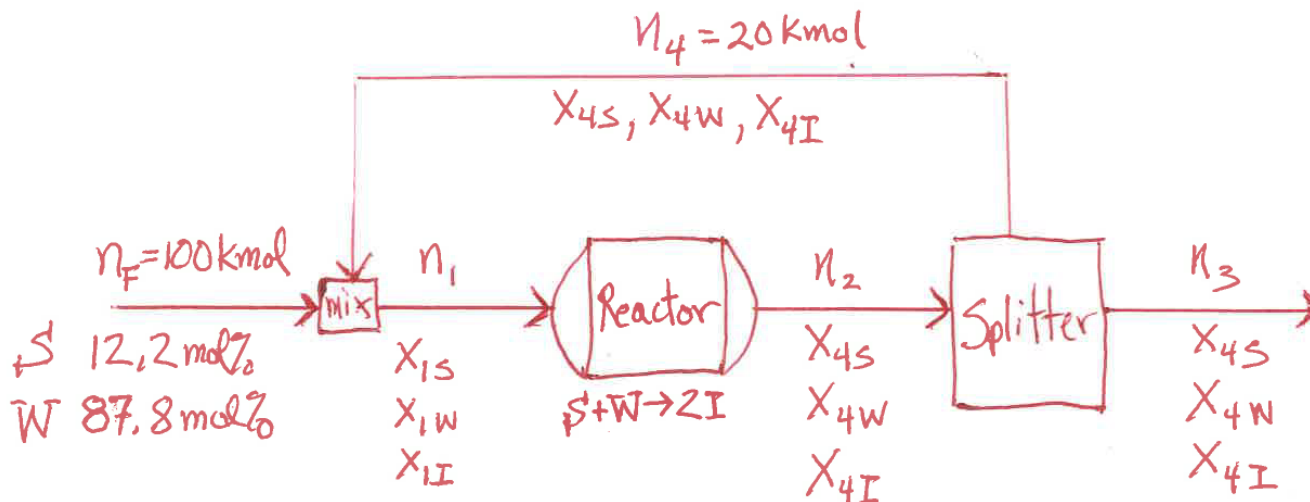
1. (50 points) Sucrose ($C_{12}H_{22}O_{11}$) can be converted to glucose and fructose (both $C_6H_{12}O_6$) according to the reaction



The mixture of glucose and fructose is called inversion sugar (I). If we give sucrose the symbol S and water the symbol W, the reaction can be written simply as



The process is shown below where the reactor has a single-pass conversion of sucrose of 80%. The process is fed 100 kmol of 12.2 mol% S and the remainder W. The reactor effluent is fed into a splitter with a product stream and a recycle stream of identical composition. The recycle stream has a flow rate of $n_4 = 20$ kmol.



- (a) Do a degree of freedom analysis following page 105 of the text and show that the problem is sufficiently specified. (5 pts.)
- (b) Using balances on the total molar flow rates, calculate the molar flow rates of the reactor input, output, and the product stream. (15 pts.)

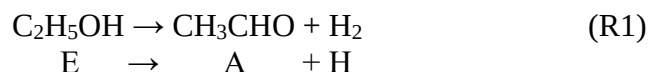
$n_1 =$ _____ kmol input, $n_2 =$ _____ kmol output, $n_3 =$ _____ kmol product

- (c) Using balances on species, write six more independent equations to calculate the mole fractions of the input and output stream for the reactor. These equations should be linear in the unknowns. (30 pts.)

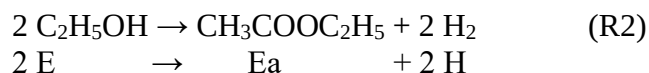
$X_{1S} =$ _____, $X_{1W} =$ _____, $X_{1I} =$ _____

$X_{4S} =$ _____, $X_{4W} =$ _____, $X_{4I} =$ _____

2. (65 points) Acetaldehyde (CH_3CHO) is produced by dehydrogenation of ethanol ($\text{C}_2\text{H}_5\text{OH}$).



An undesired side reaction produces ethyl acetate ($\text{CH}_3\text{COOC}_2\text{H}_5$) according to the reaction



The steady molar flow rate into the reactor is 120 mol/s of pure ethanol gas at 200°C and 1 atm. The total molar flow rate of the stream exiting the reactor is 166 mol/s containing ethanol (E), hydrogen (H), acetaldehyde (A), and ethyl acetate (Ea) also at 200°C and 1 atm. All components of the exit stream are gases. To maintain this isothermal condition, 2228 kJ/s of heat is added to the reactor.

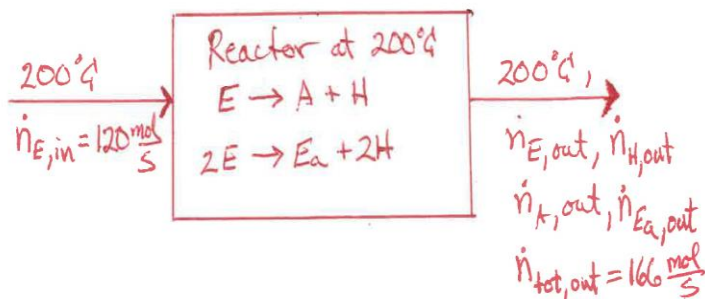
Data:

Ethanol: $C_p(\text{J/mol K}) = 21.96 + 0.1477 T(\text{K})$

Acetaldehyde: $C_p(\text{J/mol K}) = 61.00$

Hydrogen: $C_p(\text{J/mol K}) = 29.20$

Ethyl Acetate: $C_p(\text{J/mol K}) = 42.59 + 0.2397 T(\text{K})$



- (a) Write species balances on the reactor in terms of the extents of reaction for both reactions, that is, ξ_1 and ξ_2 . Also give the change in total moles over the reactor in terms of ξ_1 and ξ_2 . (15 pts.)
- (b) Calculate the enthalpy of reaction for both reaction as written using enthalpies of formation from the textbook. (10 pts.)

$$\Delta \hat{H}_{R1}^\circ = \text{_____ kJ/mol}, \quad \Delta \hat{H}_{R2}^\circ = \text{_____ kJ/mol}$$

- (c) Write an energy balance on the reactor ignoring shaft work and mechanical energy. Combine the resulting energy balance with the result from Part (a) for the total mole balance to determine ξ_1 and ξ_2 for both reactions. Use the heat capacity data given above rather than in the text. (20 pts)

$$\xi_1 = \text{_____ mol/s}, \quad \xi_2 = \text{_____ mol/s}$$

- (d) Calculate the species molar flow rates of the exiting stream from the reactor in mol/s. Complete the table below including the enthalpies at 200 °C. (10 pts.)

Species	$\dot{n}_{exit}$ (mol/s)	$\hat{H}(200^\circ\text{C})$
Ethanol (E)		
Acetaldehyde (A)		
Hydrogen (H)		
Ethyl Acetate (Ea)		

- (e) Calculate the fractional conversion of ethanol and the fractional yield of acetaldehyde. (10 pts.)

$$f_{C, \text{Ethanol}} = \text{_____}, \quad \text{fractional yield} = \text{_____}$$